

GridBalance — Data Export — Gridbalance Claims Audit

Official Research Data Export — Total Records: 6

claim	source	evidence	supported	risk_level	recommended_neutral_wording
Perfect classification accuracy (F1=1.00) on SGCC dataset	sgcc_baseline_results.csv (legacy benchmark)	REAL_SGCC_BASELINE_RESULTS.csv demonstrates real SGCC F1 is 0.401-0.416, not 1.00.	Unsupported (Legacy Synthetic Artifact)	High	The legacy synthetic dataset produced artificial F1=1.00 due to deterministic zero injection. Real SGCC evaluation yields Test F1 = 0.4014 - 0.4161.
The 18-feature model is universally superior to the 42-feature model	SGCC_SHAP_FEATURE_ANALYSIS.md	Full 42-feature model achieves Test F1 = 0.4161 vs 18-feature Test F1 = 0.4014.	Partially Supported	Medium	The 18-feature model offers comparable performance (Test F1 = 0.4014 vs 0.4161) with 57% fewer features and improved noise resilience, presenting a performance-complexity tradeoff.
Missingness features prove meter tampering	SGCC_SHAP_FEATURE_ANALYSIS.md	Missingness features (missing_streak_count, missing_count) rank #1 in SHAP importance, but missingness reflects telecommunication drops as well as physical tampering.	Unsupported (Causal Misinterpretation)	High	Missingness features exhibit strong predictive association (SHAP importance) with theft flags without establishing direct physical causality.
GridBalance is a production-ready smart grid system	General project branding	Evaluated on historical datasets without live utility deployment, operational cost calibration, or concept drift monitoring.	Unsupported	Medium	GridBalance is a research-grade prototype demonstration integrating frozen ML models with a FastAPI inference engine and React dashboard.
UCI Load Forecaster achieves 0.9944 R^2 and 1.93% sMAPE	UCI_ERROR_ANALYSIS.md & UCI_MODEL_COMPARISON.csv	Sealed test evaluation on H2 2014 load data confirms MAE = 4,421.55 kWh, RMSE = 6,654.94 kWh, R^2 = 0.9944, sMAPE = 1.93%.	Fully Supported	Low	The frozen XGBoost regressor achieves an R^2 of 0.9944 and sMAPE of 1.93% on the sealed 6-month aggregate load test set.
Synthetic evasion experiments evaluate real-world physical meter bypass	SGCC_ROBUSTNESS_ANALYSIS.md	Evasion experiments simulate mathematical load scaling (x'_t = alpha * x_t), which does not capture all physical bypass hardware interventions.	Partially Supported	Medium	Controlled evasion simulations evaluate model sensitivity to multiplicative load scaling, representing a mathematical approximation of theft evasion.